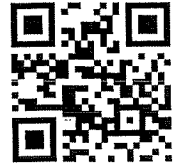


Sample Solutions

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Q1. Determine whether the following statements are true or false and give justification for your answers: If a statement is true, prove it. If a statement is false, either explain why it is false or give an example demonstrating why it is false.

(a) (5 points) If a real vector space has more than one element, then it is infinite.

TRUE Suppose that $\vec{v} \in V$.
We have $c\vec{v} \in V$ for all $c \in \mathbb{R}$.
The set of reals is infinite and
so V is an infinite set.

(b) (5 points) Every element of a vector space has at most one additive inverse.

TRUE Every element of $\vec{v} \in V$
has at most one additive inverse.

In fact, all elements have
exactly one additive inverse
because additive inverses
are unique.



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Q2. Consider the set of matrices

$$W = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : a+b=c+d \text{ and } a+c=b+d \right\}$$

(a) (4 points) Prove that W is a vector subspace of $M_{2 \times 2}(\mathbb{R})$.We pick $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\begin{pmatrix} p & q \\ r & s \end{pmatrix} \in W$.

The linear combo

$$k \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} p & q \\ r & s \end{pmatrix} = \begin{pmatrix} ak+p & bk+q \\ ck+r & dk+s \end{pmatrix}$$

satisfies:

$$(ak+p) + (bk+q) = (ck+r) + (dk+s)$$

$$(ak+p) + (ck+r) = (bk+q) + (dk+s)$$

because $a+b=c+d$ and $a+c=b+d$.(b) (3 points) Find a basis α for W .

consider

$$\alpha = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right\}$$

We check that α is lin. indep

$$\text{If } x_1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + x_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

$$\text{then } \begin{pmatrix} x_1 & x_2 \\ x_2 & x_1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \text{ and so } x_1 = x_2 = 0$$

It follows α is indep.To see that $\text{span}(\alpha) = W$ see pg. 11



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(c) (3 points) Express the matrix

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

in the basis α that you found in Q2(b).

We have:

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 1 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + 1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$



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Q3. Let V be a vector space and let $v, w \in V$. Prove that the set $\{v, w\}$ is linearly independent if and only if the set $\{v+w, v-w\}$ is linearly independent.

$\Rightarrow$ Suppose $\{\vec{v}, \vec{w}\}$ is indep.

consider

$$x_1(\vec{v} + \vec{w}) + x_2(\vec{v} - \vec{w}) = \vec{0}$$

$$\Leftrightarrow (x_1 + x_2)\vec{v} + (x_1 - x_2)\vec{w} = \vec{0}$$

By the independence of $\{v, w\}$ this gives:

$$\begin{cases} x_1 + x_2 = 0 \\ x_1 - x_2 = 0 \end{cases} \Leftrightarrow \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & -1 & 0 \end{array} \right]$$

We solve the homogeneous system

$$\left[\begin{array}{cc|c} 1 & 1 & 0 \\ 1 & -1 & 0 \end{array} \right] \xrightarrow{(-1)R_1 + R_2} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & -2 & 0 \end{array} \right] \xrightarrow{-\frac{1}{2}R_2} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{(-1)R_2 + R_1} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \text{ Thus, } x_1 = x_2 = 0 \text{ and } \{\vec{v} + \vec{w}, \vec{v} - \vec{w}\} \text{ is indep.}$$

$\Leftarrow$ Suppose $\{\vec{v} + \vec{w}, \vec{v} - \vec{w}\}$ is indep.

We have:

$$x_1\vec{v} + x_2\vec{w} = \left(\frac{x_1 + x_2}{2}\right)(\vec{v} + \vec{w}) + \left(\frac{x_1 - x_2}{2}\right)(\vec{v} - \vec{w})$$

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Q4. Find two distinct polynomials of degree three such that $f(-1) = f(0) = 1$ and $f(1) = 3$.

consider $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3$.

We have:

$$\begin{cases} 1 = f(-1) = a_0 - a_1 + a_2 - a_3 \\ 1 = f(0) = a_0 \\ 3 = f(1) = a_0 + a_1 + a_2 + a_3 \end{cases}$$

This gives a linear system:

$$\left[\begin{array}{cccc|c} 1 & -1 & 1 & -1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 3 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_1} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 1 & -1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 3 \end{array} \right]$$

$$\xrightarrow{(-1)R_1 + R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 \\ 1 & 1 & 1 & 1 & 3 \end{array} \right] \xrightarrow{(-1)R_1 + R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 1 & 1 & 1 & 2 \end{array} \right]$$

$$\xrightarrow{R_2 + R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 & 2 \end{array} \right] \xrightarrow{(-1)R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 2 \end{array} \right]$$

$$\xrightarrow{\frac{1}{2}R_3} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_3 + R_2} \left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array} \right]$$

We obtain: $a_0 = 1$, $a_1 + a_3 = 1$, $a_2 = 1$.

Thus, we may pick:

$$f(x) = 1 + x^2 + x^3 \quad \text{and}$$

$$f(x) = 1 - x + x^2 + 2x^3$$



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Q5. Determine whether the following statements are true or false and give justification for your answers: If a statement is true, prove it. If a statement is false, either explain why it is false or give an example demonstrating why it is false.

- (a) (5 points) If V and W are vector spaces, and $T: V \rightarrow W$ is a function such that $T(0_V) = 0_W$ then T is a linear transformation.

FALSE Consider $T: \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = x^2$.
We have $T(0_{\mathbb{R}}) = 0_{\mathbb{R}}$ but T is
NOT linear because:

$$4 = 2^2 = T(1+1) \neq T(1) + T(1) = 2.$$

- (b) (5 points) Let V be a two-dimensional vector space with bases α and β . The matrix representation of the zero transformation $T: V \rightarrow V$ given by $T(v) = 0$ for all $v \in V$ is $[T]_{\alpha}^{\beta} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$.

TRUE Pick bases $\alpha = \{\vec{v}_1, \vec{v}_2\}$ and $\beta = \{\vec{w}_1, \vec{w}_2\}$
We have

$$T(\vec{v}_1) = \vec{0} = 0\vec{w}_1 + 0\vec{w}_2$$

$$T(\vec{v}_2) = \vec{0} = 0\vec{w}_1 + 0\vec{w}_2$$

Thus,

$$[T]_{\alpha}^{\beta} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$



Q6. Let $S = \{v_1, \dots, v_n\}$ be a linearly independent subset of $\mathbb{R}^n$. Prove that $\text{span}(S) = \mathbb{R}^n$.

First note that $\text{span}(S) \subseteq \mathbb{R}^n$ because $S \subseteq \mathbb{R}^n$.
We now check $\mathbb{R}^n \subseteq \text{span}(S)$.

Pick $\vec{v} \in \mathbb{R}^n$. We check $\vec{v} \in \text{span}\{\vec{v}_1, \dots, \vec{v}_n\}$.

Consider

$$a_1 \vec{v}_1 + a_2 \vec{v}_2 + \dots + a_n \vec{v}_n + a_{n+1} \vec{v} = 0$$

writing this in coordinates, we get a homo.
linear system with more variables
($n+1$) than equations (n). |

Thus, the system has a non-trivial solu.

If $a_{n+1} = 0$ then we get a non-trivial
dependence: $a_1 \vec{v}_1 + a_2 \vec{v}_2 + \dots + a_n \vec{v}_n = 0$

This contradicts the independence of S .

If $a_{n+1} \neq 0$ then we can write:

$$\vec{v} = \frac{1}{a_{n+1}} (-a_1 \vec{v}_1 - \dots - a_n \vec{v}_n)$$

This expresses $\vec{v} \in \text{span}(S)$.

Thus, $\mathbb{R}^n \subseteq \text{span}(S)$.



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Q7. Let $\alpha = \{\sin(x), \cos(x)\}$ and $V = \text{span}(\alpha)$. Let $D: V \rightarrow V$ be the derivative linear map $D(f(x)) = f'(x)$. Find a basis β for V such that

$$[D]_{\alpha}^{\beta} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

$$\begin{aligned} \text{We have } D(\sin(x)) &= \cos(x) \\ D(\cos(x)) &= -\sin(x). \end{aligned}$$

Thus, we pick a basis:

$$\beta = \{\cos(x), -\sin(x)\}.$$

This gives:

$$D(\sin(x)) = 1 \cos(x) + 0(-\sin(x))$$

$$D(\cos(x)) = 0 \cos(x) + 1(-\sin(x)).$$

$$\text{Therefore, } [D]_{\alpha}^{\beta} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$



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Q8. Find a linear transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $(1, 1) \in \ker(T)$ and $(0, 1) \in \text{image}(T)$.

$$\text{consider } T(x, y) = (0, x - y).$$

We have that:

$$T(1, 1) = (0, 1 - 1) = (0, 0)$$

$$\text{and } T(1, 0) = (0, 1).$$

Thus, $(1, 1) \in \ker(T)$ and $(0, 1) \in \text{image}(T)$.

We check that T is linear.

$$T(c\vec{x} + \vec{y}) = T(cx_1 + y_1, cx_2 + y_2)$$

$$= (0, (cx_1 + y_1) - (cx_2 + y_2))$$

$$= c(0, x_1 - x_2) + (0, y_1 - y_2)$$

$$= cT(\vec{x}) + T(\vec{y})$$

Thus, T is linear.



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This page is for additional graded work.

If any work on this page is to be graded, indicate this **CLEARLY** on the question page.

Consider the condition $a+b=c+d$ and $a+c=b+d$.

This gives a linear system

$$\begin{cases} a+b-c-d=0 \\ a-b+c-d=0 \end{cases} \Leftrightarrow \begin{bmatrix} 1 & 1 & -1 & -1 & | & 0 \\ 1 & -1 & 1 & -1 & | & 0 \end{bmatrix}$$

$$\xrightarrow{(-1)R_1+R_2} \begin{bmatrix} 1 & 1 & -1 & -1 & | & 0 \\ 0 & -2 & 2 & 0 & | & 0 \end{bmatrix} \xrightarrow{-\frac{1}{2}R_2} \begin{bmatrix} 1 & 1 & -1 & -1 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \end{bmatrix}$$

$$\xrightarrow{(-1)R_2+R_1} \begin{bmatrix} 1 & 0 & 0 & -1 & | & 0 \\ 0 & 1 & -1 & 0 & | & 0 \end{bmatrix} \Leftrightarrow \begin{cases} a-d=0 \\ b-c=0 \end{cases}$$

Thus, $a=d$ and $b=c$. It follows,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a & b \\ b & a \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Therefore, $\text{span}\left\{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\right\} = W$.



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This page is for additional graded work.

If any work on this page is to be graded, indicate this **CLEARLY** on the question page.

We have: $x_1 \vec{v} + x_2 \vec{w} = \left(\frac{x_1 + x_2}{2} \right) (\vec{v} + \vec{w}) + \left(\frac{x_1 - x_2}{2} \right) (\vec{v} - \vec{w})$

If $x_1 \vec{v} + x_2 \vec{w} = \vec{0}$ then we get:

$$\left(\frac{x_1 + x_2}{2} \right) (\vec{v} + \vec{w}) + \left(\frac{x_1 - x_2}{2} \right) (\vec{v} - \vec{w}) = \vec{0}$$

This gives a linear system:

$$\begin{cases} \frac{1}{2}x_1 + \frac{1}{2}x_2 = 0 \\ \frac{1}{2}x_1 - \frac{1}{2}x_2 = 0 \end{cases} \iff \left[\begin{array}{cc|c} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & -\frac{1}{2} & 0 \end{array} \right]$$

$$\xrightarrow{(-1)R_1 + R_2} \left[\begin{array}{cc|c} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -1 & 0 \end{array} \right] \xrightarrow{(-1)R_2} \left[\begin{array}{cc|c} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{2R_1} \left[\begin{array}{cc|c} 1 & 1 & 0 \\ 0 & 1 & 0 \end{array} \right]$$

$$\xrightarrow{(-1)R_2 + R_1} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \text{ Thus } x_1 = x_2 = 0$$

It follows that $\{\vec{v}, \vec{w}\}$ is indep.

This is the end of the term test.